

# Software Requirements Specification for Sahayatri

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## **Verified Elder-Care & Remote Family Monitoring Platform**

Version 1.0

Simplified Academic SRS

Prepared from the supplied Sahayatri SRS

August 2026

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Note: In Microsoft Word, right-click the table of contents and select “Update Field” if page numbers need refreshing.

## Revision History

| Name                             | Date        | Reason for Changes                                                | Version |
|----------------------------------|-------------|-------------------------------------------------------------------|---------|
| Sahayatri Core Architecture Team | July 2026   | Initial baseline engineering specification                        | 1.0     |
| Reworked SRS                     | August 2026 | Simplified and reorganized using the supplied IEEE-style template | 1.0-S   |

## 1. Introduction

### 1.1 Purpose

This SRS describes the main functional and non-functional requirements of Sahayatri, a platform that connects Nepali families living abroad with verified caregivers who visit and monitor elderly family members in Nepal. It covers the Family, Caregiver, Elder Companion and Core Platform parts of the RC-1.0 release.

### 1.2 Document Conventions

Requirements are grouped as functional, security, performance, interface, data, offline and compliance requirements. P0 means essential for the release, P1 means high priority, and P2 means deferred or future work. The words MUST and SHOULD indicate requirement strength.

### 1.3 Intended Audience and Reading Suggestions

This document is intended for project developers, testers, administrators, reviewers and other project stakeholders. Readers should begin with Sections 1 and 2, then review the interfaces and system features in Sections 3 and 4, followed by non-functional requirements.

### 1.4 Product Scope

Sahayatri provides verified home-care visits, subscription and billing, proof-of-visit records, emergency SOS coordination, remote family monitoring and trust/safety support. The main goal is to make care provided to elderly parents more reliable and verifiable for family members living abroad.

- Verified caregiver onboarding and background checking.
- Care-plan subscriptions and foreign-currency billing with caregiver payouts in NPR.
- Scheduled visits with GPS check-in/out, photographs and wellness information.
- Emergency SOS alerts and dispatch coordination.
- Family dashboard with visit history, notes and manually recorded vital information.
- Ratings, complaints, missed-visit handling and dispute support.

RC-1.0 does not include a telemedicine marketplace, national HMIS integration, or automatic wearable vital-sign ingestion. These are future or out-of-scope functions.

### 1.5 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.
- IEEE Std 29148-2018 — Systems and software engineering requirements guidance.
- Senior Citizens Act, 2063 (2006) and related regulations.
- Public Health Service Act, 2075 (2018).
- Nepal Rastra Bank foreign-exchange and payment-related requirements.
- Individual Privacy Act, 2075 (2018).
- Electronic Transactions Act, 2063 (2006).
- Relevant payment, mapping, communication and security provider documentation used by the original SRS.

## 2. Overall Description

### 2.1 Product Perspective

Sahayatri is a new platform consisting of three mobile clients and a backend platform. The mobile clients serve family payers, caregivers and elderly recipients. The backend manages users, subscriptions, visits, emergency events, payments, notifications and stored evidence.

- Family App — subscriptions, caregiver browsing, visit records, SOS, calls and billing.
- Caregiver App — assigned visits, GPS check-in/out, proof photos, wellness data and SOS.
- Elder Companion — simplified schedule, calls and an easy SOS action.
- Core Platform — APIs, database, real-time communication, background jobs and external services.

### 2.2 Product Functions

1. Caregiver verification and onboarding.
2. Care-plan subscription, billing and caregiver payout.
3. Visit scheduling and proof-of-visit management.
4. Emergency SOS and dispatch coordination.
5. Remote family dashboard and health information tracking.
6. Trust, ratings, complaints and dispute handling.

### 2.3 User Classes and Characteristics

| User                  | Main Activities                                               | Main Needs                                  |
|-----------------------|---------------------------------------------------------------|---------------------------------------------|
| Family Payer          | Subscribe, view visits, billing, rating, SOS, calls           | Trustworthy care, simple billing and alerts |
| Elderly Recipient     | View visits, answer calls, trigger SOS                        | Simple interface, large controls and safety |
| Caregiver/Nurse       | Accept visits, check-in/out, upload proof, record vitals, SOS | Offline operation and clear care limits     |
| Verification Admin    | Review caregiver documents, approve/reject, handle disputes   | Efficient review and audit trail            |
| Emergency Coordinator | Receive SOS, view location, coordinate help                   | Fast alerts and clear escalation workflow   |
| Finance/Ops Admin     | Manage refunds and caregiver payout rules                     | Accurate financial records and controls     |

### 2.4 Operating Environment

- Mobile applications: React Native, targeting Android and iOS; Elder Companion is Android-first.
- Backend: Node.js/Express REST API with real-time communication support.
- Database: PostgreSQL with PostGIS for location-related operations.
- Offline storage: local SQLite-based storage for queued mobile data.
- Notifications: push notifications with SMS/voice fallback for critical alerts.
- The system must support low-bandwidth and intermittent-connectivity conditions common in rural Nepal.

### 2.5 Design and Implementation Constraints

- Caregivers must be verified before becoming bookable.
- Care recipient locations should be stored using a GPS point because written addresses may be imprecise.
- Core visit and SOS functions must continue to work during temporary network loss.
- Foreign-currency family billing and NPR caregiver payouts are treated as separate financial records.
- Caregiver tasks must respect their training and licensure level.
- Emergency coordination does not replace direct emergency-service calls when immediate life-saving action is required.

## 2.6 User Documentation

- In-app onboarding guidance for the three client applications.
- Caregiver training material covering visits, SOS and elder-abuse reporting.
- Family help/FAQ material covering subscriptions, refunds and proof of visits.
- Technical API documentation for authorized engineering use.

## 2.7 Assumptions and Dependencies

- Payment providers support the required currencies and payout operations.
- Caregiver phones provide working GPS hardware.
- Notification, mapping, video-call and payment providers remain available.
- Local emergency and healthcare services are external dependencies and their response capability varies by location.

# 3. External Interface Requirements

## 3.1 User Interfaces

- Family interface should provide simple access to subscriptions, visits, billing, alerts and care information.
- Caregiver interface should clearly show assigned visits, check-in/out, proof capture and SOS.
- Elder Companion should use large text, large touch targets, high contrast and a persistent SOS control.
- The applications should support English and Nepali.
- The system should provide clear messages for invalid GPS, failed payments and other common errors.

## 3.2 Hardware Interfaces

- Camera — used by caregivers to capture proof-of-visit photographs.
- GPS/GNSS — used for visit location validation and emergency location information.
- Optional BLE SOS pendant — may trigger the same SOS process as the in-app SOS button.
- Mobile local storage — used for offline queued records.

## 3.3 Software Interfaces

| External Service     | Purpose                                             |
|----------------------|-----------------------------------------------------|
| Stripe / PayPal      | Family subscription and foreign-currency billing    |
| eSewa / Khalti       | Caregiver payout in NPR                             |
| Twilio               | SMS/voice fallback and OTP                          |
| Agora.io             | Family and emergency video/voice calls              |
| Google Maps Platform | Maps, location display and distance/ETA support     |
| PostgreSQL/PostGIS   | Application data and geographic validation          |
| Cloud media storage  | Proof photos and selected emergency-call recordings |

## 3.4 Communications Interfaces

- Client-server communication must use secure HTTPS.
- Real-time communication must use secure WebSockets.
- Push notifications should use platform notification services.
- SOS alerts should have a fallback path using SMS/voice when normal delivery is not acknowledged.
- The REST API should use versioned endpoints.

## 4. System Features

### 4.1 Caregiver Verification and Onboarding

Only verified caregivers can accept paid visits. The process includes document submission, administrative review, interview, training and activation.

#### Stimulus/Response Sequences

| Stimulus                        | System Response                                                        |
|---------------------------------|------------------------------------------------------------------------|
| Applicant submits phone number  | System verifies the phone using OTP and starts the onboarding process. |
| Applicant submits ID and selfie | System stores the documents and sends the application for review.      |
| Admin reviews application       | System allows approval, rejection or request for further review.       |
| Interview is passed             | System makes the training step available.                              |
| Training is completed           | Caregiver becomes active and bookable.                                 |

#### Functional Requirements

**REQ-FUNC-001 [P0]:** Caregiver onboarding **MUST** require identity documents and a live selfie before activation.

**REQ-FUNC-002 [P0]:** A caregiver **MUST NOT** receive visits or appear in family search until approved.

**REQ-FUNC-003 [P1]:** The system **MUST** record verification status changes and the reason for each change.

**REQ-FUNC-004 [P1]:** The system **SHOULD** verify declared caregiver training or licence level before assigning restricted tasks.

**REQ-FUNC-005 [P0]:** An elder-abuse or neglect report **MUST** immediately block the caregiver from new assignments pending review.

### 4.2 Subscription and Billing

Family payers can subscribe to care plans and pay in supported foreign currencies. Completed visits create caregiver payout records in NPR.

#### Stimulus/Response Sequences

| Stimulus                   | System Response                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------|
| Family selects a care plan | System creates a subscription and waits for successful payment.                            |
| Payment succeeds           | Subscription becomes active and visit scheduling is enabled.                               |
| Visit is completed         | System creates a payout item subject to the configured dispute/holding period.             |
| Payment fails              | System notifies the payer and may suspend future scheduling after the grace period.        |
| Family cancels             | Future visits are stopped after the paid period while historical records remain available. |

#### Functional Requirements

**REQ-FUNC-101 [P0]:** The system **MUST** support recurring care-plan subscriptions.

**REQ-FUNC-102 [P0]:** The system **MUST** support USD, AUD, GBP and EUR at RC-1.0 launch.

**REQ-FUNC-103 [P0]:** Raw payment-card data **MUST NOT** be stored by Sahayatri.

**REQ-FUNC-104 [P1]:** A completed, non-disputed visit **SHOULD** generate a caregiver payout item.

**REQ-FUNC-105 [P1]:** Failed payouts **SHOULD** be retried or sent for administrative review without blocking unrelated payouts.

### 4.3 Visit Scheduling and Proof-of-Visit

Caregivers complete scheduled visits using GPS check-in/out, photographs and a wellness questionnaire. The family can later view the visit evidence.

#### Stimulus/Response Sequences

| Stimulus                            | System Response                                                                |
|-------------------------------------|--------------------------------------------------------------------------------|
| Caregiver receives assignment       | System shows visit date, time and recipient information.                       |
| Caregiver checks in                 | System validates the GPS position against the recipient's registered location. |
| Caregiver records visit information | System stores questionnaire answers, notes and proof photo.                    |
| Caregiver checks out                | System records the end time and completes the visit.                           |
| Network is unavailable              | Core visit data is queued locally and synchronized when connectivity returns.  |

#### Functional Requirements

**REQ-FUNC-201 [P0]:** The system **MUST** validate caregiver check-in/out using GPS and a configurable geo-fence.

**REQ-FUNC-202 [P0]:** Proof-of-visit **MUST** include a timestamped photograph and visit information.

**REQ-OFFLINE-001 [P0]:** Core visit functions **MUST** support offline operation and later synchronization.

**REQ-FUNC-203 [P1]:** The family **MUST** be able to view completed visit records and evidence.

**REQ-FUNC-204 [P1]:** Missed visits **MUST** notify the family and restore the applicable visit credit according to business rules.

### 4.4 Emergency SOS and Dispatch

The system provides an SOS process for elderly recipients, caregivers and family users. Alerts are sent to appropriate people and can be escalated to an Emergency Coordinator.

#### Stimulus/Response Sequences

| Stimulus                   | System Response                                                           |
|----------------------------|---------------------------------------------------------------------------|
| SOS is triggered           | System creates an emergency alert and notifies linked users/coordinators. |
| Alert is acknowledged      | System records the acknowledgement and active response.                   |
| Alert is not acknowledged  | System uses SMS/voice fallback and further escalation.                    |
| Coordinator receives alert | Coordinator can view available location information and coordinate help.  |
| Emergency is resolved      | System stores the incident timeline for later review.                     |

#### Functional Requirements

**REQ-FUNC-301 [P0]:** Users **MUST** be able to trigger an SOS from supported client interfaces.

**REQ-FUNC-302 [P0]:** SOS alerts **MUST** be delivered through more than one communication channel.

**REQ-FUNC-303 [P0]:** The system **MUST** record the alert, acknowledgement and resolution history.

**REQ-FUNC-304 [P1]:** The system **SHOULD** identify the nearest available caregiver or relevant emergency facility.

**REQ-FUNC-305 [P1]:** The system **MUST** clearly state that Sahayatri coordinates emergency response and does not operate ambulances.

### 4.5 Family Dashboard, Health Information and Trust

Family users can monitor their parent's care through visit history, notes, photographs and manually recorded vitals. Ratings and complaints support service quality.

## Stimulus/Response Sequences

| Stimulus                              | System Response                                                     |
|---------------------------------------|---------------------------------------------------------------------|
| Family opens dashboard                | System displays upcoming visits and recent care information.        |
| Family opens a completed visit        | System displays the visit details, evidence and caregiver notes.    |
| Caregiver records vitals              | System stores manually entered readings for later viewing.          |
| Family rates a visit/caregiver        | System stores the rating and sends low ratings to the review queue. |
| Family or caregiver reports a problem | System creates a case for administrative review.                    |

## Functional Requirements

**REQ-FUNC-401 [P1]:** The family **MUST** be able to view scheduled and completed visits.

**REQ-FUNC-402 [P1]:** Manually recorded vital information **MUST** be clearly identified as caregiver-entered data.

**REQ-FUNC-403 [P1]:** The system **SHOULD** show visit adherence for the current care period.

**REQ-FUNC-404 [P1]:** The system **MUST** support ratings and reviews for completed services.

**REQ-FUNC-405 [P1]:** Low ratings and safety complaints **MUST** be available to the appropriate administrative review process.

## 5. Other Nonfunctional Requirements

### 5.1 Performance Requirements

- Standard REST read requests should target a p95 response time below 500 ms under normal server conditions.
- Geo-location write operations should target a p95 response time below 1200 ms, excluding rural mobile network delay.
- SOS server-side notification fan-out should be completed quickly enough to support near-real-time emergency handling.
- Offline queued records should synchronize efficiently after connectivity returns.
- Payout processing should support batch execution without one failed payout blocking other successful payouts.

### 5.2 Safety Requirements

- Unacknowledged SOS alerts **MUST** have a fallback escalation process.
- Suspending a caregiver during an active visit **MUST** trigger a welfare/safety check rather than silently ending the visit.
- Caregiver training **MUST** include basic emergency response guidance and when to contact local emergency services directly.
- Sahayatri's SOS function is a coordination and notification service, not a replacement for emergency services.

### 5.3 Security Requirements

- Sensitive stored data **MUST** be encrypted.
- Authentication **MUST** use secure short-lived access tokens and protected refresh mechanisms.
- Role-Based Access Control (RBAC) **MUST** restrict each user to permitted information and actions.
- Payment processing **MUST** avoid handling raw card information.
- Administrative access to sensitive personal information **MUST** be recorded in an audit log.

## 5.4 Software Quality Attributes

| Attribute       | Requirement                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------|
| Availability    | Critical services should remain available and support recovery from failures.                  |
| Usability       | The Elder Companion interface should be simple enough for elderly first-time users.            |
| Reliability     | Offline visit records should not be silently lost when connectivity is interrupted.            |
| Maintainability | Business logic should be modular and tested.                                                   |
| Adaptability    | External payment and communication providers should be replaceable through service interfaces. |

## 5.5 Business Rules

- Subscription cancellation normally takes effect at the end of the paid period.
- Missed visits should notify the family and restore the relevant visit credit.
- Caregiver payouts are released according to the configured holding/dispute period and payout schedule.
- Caregiver rate changes are controlled by authorized operations staff.
- Caregiver tasks must follow the caregiver's verified training/licensure level.
- Elder-abuse or neglect reports receive higher priority than ordinary service disputes.

## 6. Other Requirements

### 6.1 Database Requirements

The database should maintain the main entities needed for family accounts, payers, subscriptions, care recipients, caregivers, visit schedules, visit logs, proof photos, vitals, emergency alerts, payouts and status/audit records.

- A care recipient may have multiple visit schedules and visit records.
- A caregiver may complete many visits and receive many payout records.
- A completed visit may have proof photographs and manually entered vitals.
- Emergency alerts must keep an event timeline.
- Caregiver onboarding status changes must be auditable.
- Money values should be stored using integer minor units to avoid floating-point rounding errors.

### 6.2 Data Retention

| Data                                    | Retention in Original SRS                  |
|-----------------------------------------|--------------------------------------------|
| Proof-of-visit photos and visit records | Minimum 5 years                            |
| Emergency alert timelines               | Indefinite                                 |
| Payment and payout records              | Minimum 7 years                            |
| Caregiver KYC documents                 | Active status + 3 years after deactivation |
| General chat/call metadata              | 1 year                                     |
| SOS-triggered call recordings           | 5-year proof-of-visit-equivalent period    |

### 6.3 Privacy and Compliance

- Consent must be obtained and presented in clear language, including Nepali where applicable.
- The system should collect only information necessary for care, safety, billing and platform operation.
- Caregiver conduct and reporting workflows must follow applicable Nepalese requirements.
- Cross-border processing of relevant personal information must be documented and controlled.

## Appendix A: Glossary

| Term                  | Meaning                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------|
| Diaspora              | Nepali citizens living and working outside Nepal who use Sahayatri to arrange care.        |
| Proof-of-Visit        | GPS-validated visit record with timestamp, photo and care information.                     |
| Geo-fence             | Configured geographic boundary around a care recipient's registered home location.         |
| KYC                   | Know Your Customer / identity verification process.                                        |
| SOS                   | Emergency alert and escalation process.                                                    |
| PostGIS               | PostgreSQL extension used for geographic data and spatial queries.                         |
| ANM                   | Auxiliary Nurse Midwife.                                                                   |
| CMA                   | Community Medical Assistant.                                                               |
| Escrow/holding period | Time between visit completion and payout eligibility during which a dispute may be raised. |

## Appendix B: Analysis Models

### B.1 Context-Level Data Flow

Family Payer → Sahayatri ← Caregiver; Elderly Recipient → Sahayatri; Verification Admin and Emergency Coordinator → Sahayatri. Sahayatri connects to payment services, payout services, notification services, video/voice services and mapping services.

### B.2 Simplified Proof-of-Visit Flow

- Caregiver receives a scheduled visit.
- Caregiver checks in with GPS.
- System validates the location.
- Caregiver records questionnaire information and proof photo.
- Caregiver checks out.
- System stores the completed visit and makes the visit information available to the family.
- Eligible payout information is created for the caregiver.

### B.3 Simplified Entity Relationships

FamilyAccount → Payer → Subscription → CareRecipient → VisitSchedule → VisitLog. VisitLog connects to ProofPhoto and VitalsReading and may create a CaregiverPayoutLineItem. Caregiver connects to VisitLog and CaregiverStatusHistory. CareRecipient connects to EmergencyAlert.

### B.4 Use Case Summary

| Use Case                           | Primary Actor                                         |
|------------------------------------|-------------------------------------------------------|
| Register and verify caregiver      | Caregiver / Verification Admin                        |
| Subscribe to care plan             | Family Payer                                          |
| Complete scheduled visit           | Caregiver                                             |
| Trigger and resolve SOS            | Elderly Recipient / Caregiver / Emergency Coordinator |
| Monitor parent                     | Family Payer                                          |
| Receive caregiver payout           | Caregiver / Finance-Ops Admin                         |
| Review dispute or safety complaint | Verification Admin / Operations                       |

## Appendix C: To Be Determined / Future Work

- Wearable IoT integration and continuous automatic health monitoring.

15. AI-based health anomaly detection.
16. Automated national criminal-record API integration if an appropriate government service becomes available.
17. Telemedicine or doctor video-consultation marketplace.
18. National HMIS integration.
19. Additional diaspora settlement currencies beyond the RC-1.0 launch currencies.
20. Family self-service custom questionnaire authoring.
21. Formal multi-tenant agency dashboards.